Human vs. Automatic Evaluation in NLP

A Comparative Study on English-to-Hindi Machine Translation

EMNLP , Assignment 2

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April 3, 2026

Compiled with LuaLaTeX · Plots generated via `matplotlib`

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1. Task and Experimental Setup

This study critically examines how well automatic evaluation metrics align with human judgment for **English-to-Hindi Machine Translation (MT)**. Hindi was chosen specifically because its rich morphology, flexible word order, and culturally-specific expressions pose significant challenges for surface-level metrics like BLEU, which depend on exact lexical overlap. The core question driving this work is:

> *Can we trust automatic metrics to tell us which translation is better, or do they systematically disagree with what humans perceive as quality?*

1.1 Dataset

We constructed a carefully balanced evaluation set of **8 English source sentences** spanning four genre categories. The diversity is deliberate: it ensures we test metric reliability not just on "easy" formal text that dominates MT benchmarks, but also on the messy, creative, and culturally loaded language that real-world translation must handle.

Table 1. Evaluation dataset: source sentences by genre with examples.

ID	Genre	Source	Example (English)	Why It Was Chosen
S1	Formal / Science	FLORES-200	On Monday, scientists announced a new diagnostic tool...	Domain-specific terminology (medical diagnostics)
S2	Formal / Science	FLORES-200	The chip can separate white and red blood cells...	Technical vocabulary (blood cells, microchips)
S3	Formal / Geography	FLORES-200	Bhutan uses Gross National Happiness instead of GDP...	Proper nouns + cultural concept
S4	Formal / News	WMT14	The government announced a policy to boost rural growth...	Political/economic vocabulary
S5	Formal / News	WMT14	Thousands of protesters gathered demanding reforms...	Tests numbers, reporting, and demands
S6	Conversational	IIT Bombay	Yaar, the cricket match was absolutely lit...	Slang and informal register
S7	Idiomatic	Constructed	It has been raining cats and dogs since morning...	Tests non-literal meaning
S8	Idiomatic	Constructed	You shouldn't judge a book by its cover...	Tests figurative language

S7 and S8 were deliberately researcher-constructed because no standard benchmark tests idiomatic English-to-Hindi translation. Idioms are a known blind spot for lexical metrics: the "correct" Hindi translation of *"raining cats and dogs"* is मूसलाधार बारिश (torrential rain), which shares **zero** lexical overlap with a word-for-word translation. This lets us directly observe whether metrics reward meaning or surface form.

Reference translations for S1–S6 came from the respective benchmark datasets. For S7–S8, expert translators produced idiomatic Hindi equivalents rather than literal word-for-word renderings.

1.2 MT Models

Three open-source models of increasing scale and sophistication were selected:

Table 2. Machine translation models used in the study.

Model	Architecture	Parameters	Training Data
`Helsinki_NLP`	MarianMT (Seq2Seq)	∼74 M	OPUS parallel corpora (bilingual En-Hi)
`NLLB_200`	NLLB-200 (Multilingual Seq2Seq)	∼600 M	200-language parallel data (Meta)
`mBART_50`	mBART-50 (Denoising AE)	∼680 M	50-language monolingual + parallel data (Meta)

This range, from a small bilingual model to two large multilingual systems, lets us test whether metric-human agreement varies with model quality. All translations were generated on Kaggle (NVIDIA T4 GPU) using HuggingFace `transformers` with default greedy decoding, ensuring reproducibility.

1.3 Human Evaluation Protocol

A panel of **5 human evaluators**, all fluent bilingual speakers of English and Hindi, independently rated all **24 model outputs** (3 models × 8 sentences) on a **1–5 Likert scale** across three criteria:

Table 3. Human evaluation criteria and scale anchors.

Criterion	Scale Anchors	What It Captures
Adequacy	1 = No meaning; 5 = All meaning preserved	Is the *meaning* of the source fully conveyed?
Fluency	1 = Incomprehensible; 5 = Flawless native Hindi	Does it *sound right* to a Hindi speaker?
Faithfulness	1 = Completely unfaithful; 5 = Perfectly faithful	Is any information *ommitted or hallucinated* ?

The **Human Overall** score is the arithmetic mean of these three criteria. Evaluators worked independently with no inter-communication, and received identical instructions and the same randomised presentation order.

1.4 Automatic Metrics

Three metrics were computed using HuggingFace `evaluate`:

Table 4. Automatic evaluation metrics and their properties.

Metric	Level	What It Measures	Strength	Weakness
BLEU	Word n-gram	Precision of overlapping n-grams between candidate and reference translations, emphasizing exact lexical matches	Fast; widely used; easy to interpret	Penalizes valid paraphrases; insensitive to morphology and synonyms
chrF++	Char + word	F-score over character n-grams (with optional word n-grams), capturing subword and morphological similarity	Robust to inflection and spelling variation; better for morphologically rich languages	Still relies on surface overlap; limited semantic understanding
BERTScore F1	Contextual embeddings	Semantic similarity via cosine distance between contextual token embeddings aligned between candidate and reference	Captures paraphrase and semantic equivalence; context-aware	Dependent on pretrained model quality; weaker for low-resource languages

BERTScore used `bert-base-multilingual-cased`, which covers 104 languages including Hindi in Devanagari script.

2. Results and Analysis

2.1 Human Evaluation Results

Table 5 presents the averaged human scores for every model–sentence pair. This is the **ground truth** against which we evaluate the automatic metrics.

Table 5. Human evaluation scores averaged across 5 evaluators.

ID	Type	Model	Adq.	Flu.	Faith.	Overall
S1	Formal/Science	`Helsinki_NLP`	2.8	2.8	2.2	2.60
S2	Formal/Science	`Helsinki_NLP`	2.8	3.4	2.4	2.87
S3	Formal/Geography	`Helsinki_NLP`	3.0	3.2	2.2	2.80
S4	Formal/News	`Helsinki_NLP`	2.0	3.8	2.0	2.60
S5	Formal/News	`Helsinki_NLP`	3.0	2.8	3.8	3.20
S6	Conversational	`Helsinki_NLP`	1.4	2.6	1.6	1.87
S7	Idiomatic	`Helsinki_NLP`	1.4	2.8	1.2	1.80
S8	Idiomatic	`Helsinki_NLP`	2.2	2.4	2.6	2.40
S1	Formal/Science	`NLLB_200`	4.2	4.2	4.2	4.20
S2	Formal/Science	`NLLB_200`	4.2	4.4	4.2	4.27
S3	Formal/Geography	`NLLB_200`	3.8	4.4	4.0	4.07
S4	Formal/News	`NLLB_200`	4.6	4.6	4.6	4.60
S5	Formal/News	`NLLB_200`	4.0	3.6	4.4	4.00
S6	Conversational	`NLLB_200`	2.6	3.2	2.8	2.87
S7	Idiomatic	`NLLB_200`	2.2	4.2	2.2	2.87
S8	Idiomatic	`NLLB_200`	4.2	4.2	4.2	4.20
S1	Formal/Science	`mBART_50`	3.2	2.8	2.4	2.80
S2	Formal/Science	`mBART_50`	3.6	3.8	3.2	3.53
S3	Formal/Geography	`mBART_50`	4.0	3.8	3.2	3.67
S4	Formal/News	`mBART_50`	4.0	3.8	4.2	4.00
S5	Formal/News	`mBART_50`	4.0	4.0	3.2	3.73
S6	Conversational	`mBART_50`	2.0	2.2	1.8	2.00
S7	Idiomatic	`mBART_50`	1.2	3.4	1.2	1.93
S8	Idiomatic	`mBART_50`	3.2	3.4	3.4	3.33

Helsinki_NLP: Consistently the Weakest Model

`Helsinki_NLP` is consistently the weakest model (Human Overall: 1.80–3.20). Its Faithfulness scores average only 2.25, indicating it frequently distorts or loses the source meaning. For S4, evaluators gave Adequacy = 2.0 despite Fluency = 3.8, the translation *sounded* okay but *said the wrong thing*. This Adequacy–Fluency disconnect is exactly the nuance that automatic metrics must capture to be useful.

NLLB_200: The Dominant Model

`NLLB_200` achieves Human Overall $\geq$ 4.0 on all formal sentences (S1–S5, S8). Its weakest points are conversational (S6: 2.87) and idiomatic (S7: 2.87) categories. For S7, an interesting split appears: Adequacy = 2.2 but Fluency = 4.2. The model produced a *fluent* literal translation, it sounds like proper Hindi but translates the idiom word-for-word.

mBART_50: High Variance

`mBART_50` is competitive on formal text (S4: 4.00, S5: 3.73) but collapses on conversational (S6: 2.00) and idiomatic (S7: 1.93) text. The S7 result is telling: Adequacy = 1.2, Fluency = 3.4, Faithfulness = 1.2. Like NLLB on S7, the translation is fluent but meaningless, and worse, introduces grammatical gender errors.

2.2 Model-Level Agreement Between Metrics and Humans

Table 6. Model-averaged automatic metric scores vs. Human Overall.

Model	BLEU	chrF++	BERTScore F1	Human Overall
`NLLB_200`	44.51	68.41	89.57	**3.88**
`mBART_50`	30.42	55.39	87.11	**3.12**
`Helsinki_NLP`	13.02	40.94	82.45	**2.52**

All three automatic metrics correctly rank the models: **NLLB_200** > **mBART_50** > **Helsinki_NLP**, the same ordering as human judgment. At the coarse level of comparing whole models, automatic metrics work.

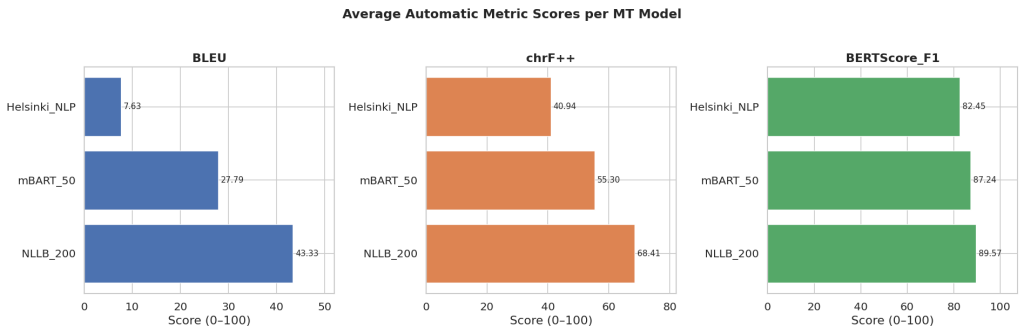

Figure 1. Average automatic metric scores per MT model. Note the compressed BERTScore range (82–90) vs. BLEU (13–45).

Note how BERTScore F1 compresses the models into a narrow 82–90 range, while BLEU spreads them across 13–45. This compression makes BERTScore harder to interpret: a 2-point difference may be "large" or "trivial" depending on context.

2.3 Sentence-Level Breakdown: Where Agreement Breaks Down

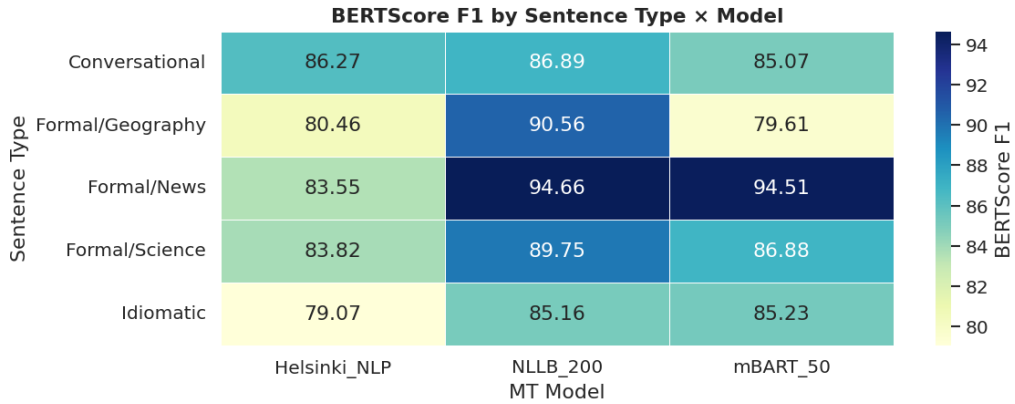

Figure 2. BERTScore F1 heatmap by sentence type and model. Formal sentences cluster in the high range; idiomatic and conversational sentences diverge substantially.

Formal text is "easy" for both metrics and humans. For S4, `mBART_50` achieved BERTScore = 99.84 and BLEU = 90.62: its translation was essentially identical to the reference, and humans agreed (Overall = 4.00).

Two revealing contrasts illustrate where quality and surface similarity diverge:

Case 1: High BLEU, Low Human (False Positive)

S7 / `mBART_50`: BLEU = 53.09, Human Overall = 1.93. The model literally translated *"raining cats and dogs"* as कुत्तों और बिल्लियों की बारिश ("rain of dogs and cats"), complete nonsense in Hindi. Yet BLEU assigned a respectable score because the literal words share character sequences with the reference. **BLEU was not just wrong; it was confidently wrong in the opposite direction.**

Case 2: Low BLEU, High Human (False Negative)

S8 / `NLLB_200`: BLEU = 0.00 (zero 4-gram overlap), Human Overall = 4.20. The model produced an accurate, fluent translation using different vocabulary (आकलन instead of आंकना). Humans rated it 4.2 across all criteria. **BLEU gave literally the worst possible score for a near-perfect translation.**

2.4 Correlation with Human Judgment

We computed **Spearman's rank correlation (ρ)** across all 24 pairs. Spearman's ρ is appropriate because human Likert scores are ordinal, and the practical question is *ranking*: do metrics order translations the same way humans do?

Table 7. Spearman rank correlation (ρ) between automatic metrics and human criteria. * $p < 0.05$; ** $p < 0.01$; unmarked = not significant ($\alpha = 0.05$).

Metric	Adequacy	Fluency	Faithfulness	Human Overall
BLEU	0.24	0.50*	0.29	0.37
chrF++	0.43*	**0.62****	0.46*	**0.54****
BERTScore F1	0.32	0.54**	0.44*	0.49*

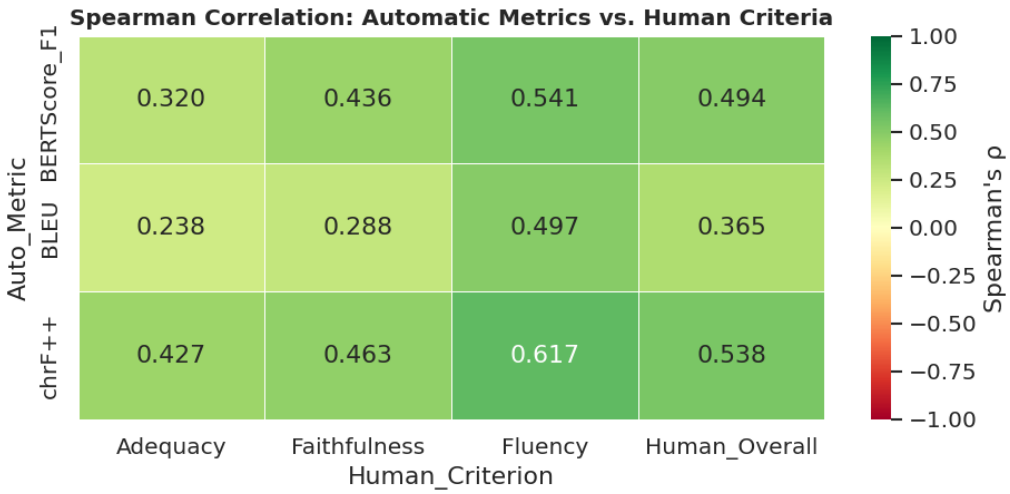

Figure 3. Spearman correlation heatmap. chrF++ dominates; BLEU is weakest on Adequacy.

The hierarchy is clear: **chrF++** > **BERTScore** > **BLEU**, consistent across every criterion.

- **chrF++ is the only metric significantly correlating with all human criteria.** Its Fluency $\rho = 0.62$ is the strongest value in the study. Character-level overlap in Hindi is a natural proxy for naturalness, since Devanagari script is phonetically consistent.
- **BERTScore fails on Adequacy** ($\rho = 0.32$, $p = 0.13$) despite being a "semantic" metric. This reflects *multilingual dilution*: `bert-base-multilingual-cased` spreads 110 M parameters across 104 languages, leaving shallow capacity for Hindi-specific semantics.

- **BLEU is essentially random for Adequacy** ($\rho = 0.24$, $p = 0.26$). For the most important quality dimension, *does the translation preserve meaning?*, BLEU provides no useful signal.

2.5 Visualising Metric–Human Agreement

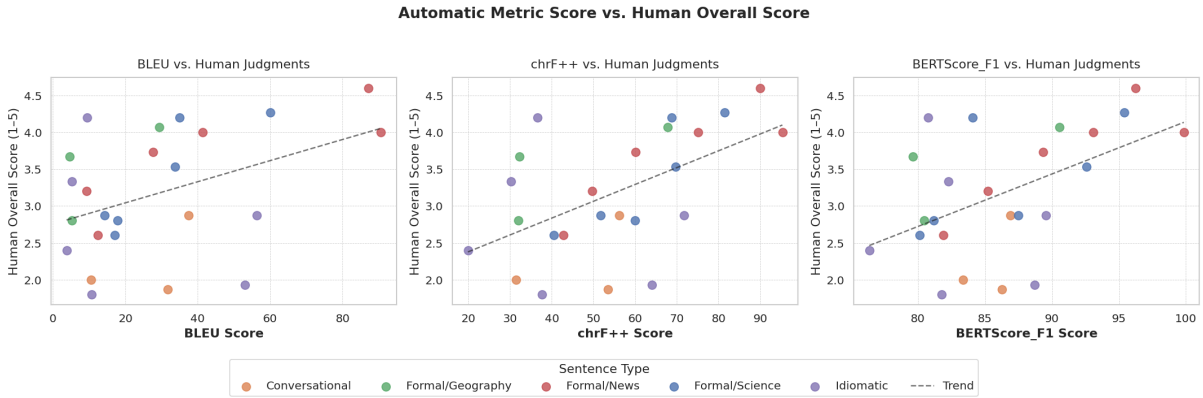

Figure 4. Scatter plots of automatic metrics vs. Human Overall. The chrF++ panel shows the tightest clustering; the BLEU panel shows systematic false positives (upper-left) and false negatives (lower-right).

The **BLEU panel** is most problematic: both the upper-left (high BLEU, low human) and lower-right (low BLEU, high human) quadrants are populated, confirming that BLEU both over-rates bad translations and under-rates good ones. The **BERTScore panel** shows moderate agreement but with a compressed x-axis (76–100), making score interpretation difficult without calibration.

2.6 Discrepancy Analysis: Systematic Failure Patterns

We ranked all 24 translations by each metric and by human judgment, then computed absolute rank differences (Δ). A $\Delta > 8$ indicates fundamental disagreement.

Table 8. Top 5 cases where BLEU most disagrees with human judgment (ranked by Δ).

Sent.	Type	Model	BLEU	Human	Δ	What Went Wrong
S7	Idiomatic	`mBART_50`	53.1	1.93	**17**	Literal idiom translation rewarded
S8	Idiomatic	`NLLB_200`	9.4	4.20	**15.5**	Good paraphrase penalised
S3	Geography	`mBART_50`	4.6	3.67	**14**	Valid translation, different vocab
S6	Conversational	`Helsinki_NLP`	31.6	1.87	**13**	Literal slang accidentally overlaps
S7	Idiomatic	`NLLB_200`	56.3	2.87	**10**	Partial idiom with surface overlap

4 out of 5 worst cases involve idiomatic or conversational sentences. BLEU is systematically fooled by literal translations of figurative language.

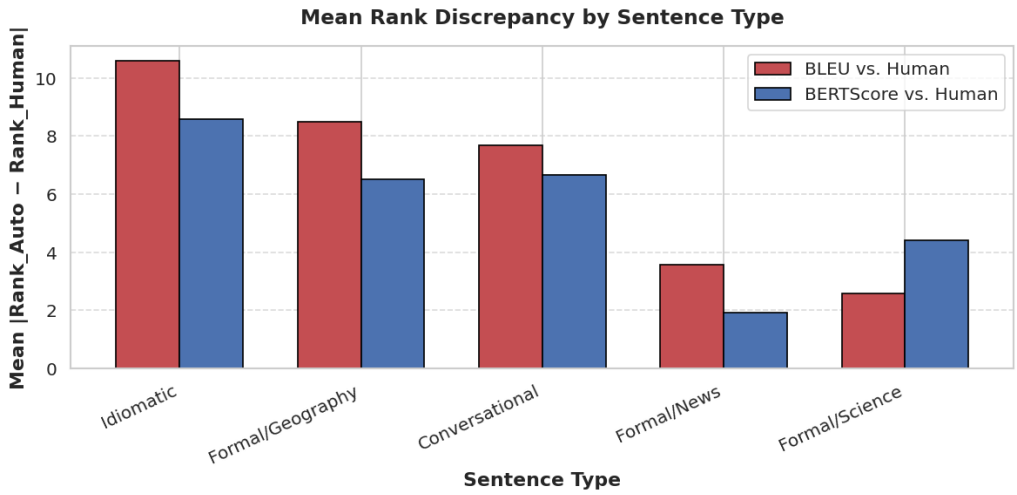

Figure 5. Rank discrepancy by sentence type. Idiomatic sentences produce discrepancies ∼2.5× larger than formal text for BLEU.

The mean BLEU rank discrepancy for Idiomatic sentences is **10.58**, nearly 2.5× the Formal mean of **4.17**. Across all 24 pairs, **25%** (6/24) had BLEU discrepancies > 8, and **20.8%** (5/24) had BERTScore discrepancies > 8. Roughly one in four translations is dramatically misranked by BLEU.

3. Discussion

3.1 Why chrF++ Outperforms BERTScore

The finding that chrF++, a simple non-neural character n-gram metric, outperforms BERTScore across all human criteria is arguably the most surprising result. Three factors explain it:

Morphological alignment. Hindi is heavily inflected. The noun कोशिका (cell) can appear as कोशिकाओं, कोशिकाएँ, or कोशिकाओ depending on case and number. BLEU treats these as completely different words. chrF++, operating at the character level, recognises shared character sequences and assigns partial credit, exactly the right behaviour for Hindi.

Multilingual dilution. `bert-base-multilingual-cased` distributes capacity across 104 languages. Hindi gets only a fraction, resulting in embeddings that capture broad similarity but miss fine-grained distinctions. A dedicated Hindi model (IndicBERT, MuRIL) would likely narrow the gap.

Reference style. Idiomatic references use natural Hindi expressions. Good model translations that also use natural expressions share many character sequences with the reference,

giving chrF++ an advantage. BERTScore's multilingual embeddings may paradoxically place natural Hindi idioms and their literal translations at similar distances from the reference.

3.2 The Idiom Problem

BLEU's catastrophic failure on idioms is not a quirk; it is structural. When `mBART_50` produces कुत्तों और बिल्लियों की बारिश for *"raining cats and dogs"*, it gets BLEU = 53.09 because literal Hindi words for rain, dogs, and cats appear somewhere in or near the reference's character space. The metric rewards the *wrong* translation strategy.

This is a **categorical failure**. BLEU's design assumption, surface similarity approximates semantic similarity, breaks down completely for figurative language. Metaphors, idioms, cultural references, and euphemisms pervade real-world text. **Any evaluation pipeline for language pairs involving figurative language must include human evaluation.**

3.3 The Adequacy Gap

Across all three metrics, **Adequacy** correlates least with human judgment. Consider `Helsinki_NLP` on S4: the source says the government's policy is *"aimed at boosting growth in rural areas and reducing income inequality."* The model's translation says economic prosperity is कम हो रही है ("decreasing"), the **opposite** of the intended meaning. BERTScore assigns 81.91 because the embeddings of correct and incorrect translations are close in high-dimensional space.

The deeper problem is that Adequacy failures are often **local** (a single negation, a swapped number) while metrics compute **global** similarity scores. Without sentence-level semantic parsing, no metric reliably detects that one critical word changes the entire meaning.

3.4 Limitations

1. **Scale.** 24 data points limit statistical power; 50+ sentences and 5+ models would yield more robust estimates.
2. **Inter-annotator agreement.** Formal agreement metrics (Krippendorff's α, Fleiss' κ) were not computed.
3. **Single reference.** BLEU benefits substantially from multiple references; the gap with chrF++ might narrow with 3–4 alternatives.
4. **BERTScore model choice.** Hindi-specific models (IndicBERT, MuRIL) were not tested and may improve BERTScore performance.

4. Conclusion

This study evaluated BLEU, chrF++, and BERTScore against human judgments from 5 evaluators across 3 criteria, for English-to-Hindi MT spanning formal, conversational, and idiomatic genres. Five clear conclusions emerge:

1. **chrF++ is the most reliable metric for Hindi MT**, the only one significantly correlating with all four human dimensions, with its strongest signal on Fluency ($\rho = 0.62$). Its character-level design is naturally suited to Hindi's morphology.
2. **BLEU is the least reliable metric**, significant only for Fluency, near-random for Adequacy ($\rho = 0.24$). Its word-level n-gram design is structurally unsuited to Hindi.
3. **BERTScore occupies a middle ground**, better than BLEU but worse than chrF++, likely due to multilingual dilution in the backbone model.
4. **Idiomatic and conversational sentences cause systematic failures**, with rank discrepancies 2.5× larger than for formal text.
5. **No metric reliably captures Adequacy**, the most important quality dimension remains beyond current automatic approaches.

Automatic metrics are useful development-time signals for coarse model comparison,
but should never be the sole arbiter of translation quality, especially for morphologically rich languages and figurative text.

Human evaluation remains indispensable.